Review questions for Lecture 3

1. What is the 23rd percentile of the following list: 6, 9, 2, 10, 50?

2. Consider the function $f(u) = (u - 3)^2 + (u - 5)^2 + (u + 4)^2 + (u - 8)^2$ What value of u minimizes f(u)?

3. Consider the function $g(u) = |u - 6| + |u - 1| + |u + 1| + |u| + |u - 4|$ What value of u minimizes g(u)?

4. Consider the following list: 6, 9, 2, 10, 50. What is the IQR? What is the MAD?

5. In what context might someone be interested in the median house price in a city? In what context might someone be interested in the mean house price in a city?

6. Consider the following list of numbers:

0.32, 0.72, 1.48, 1.50, -0.58, 0.46, -0.72, -0.31, 0.86, 0.15, 0.46, 1.87, 0.11, 0.26, -0.37, -0.99, 1.15, 0.01, -1.75, -0.51, 1.75, 0.75, -0.69, 0.40, -1.30

Use R to calculate the mean of this list.

7. Using the same list as question 6, use R to calculate the median.

8. Using the same list as question 6, use R to calculate the standard deviation.